



Akita AI

USER'S MANUAL

**AI TRAINING FRAMEWORK LEVERAGING SCALABLE
HIGH-PERFORMANCE COMPUTING (HPC) CLUSTER FOR
ACADEMIC LITERATURE RECOMMENDATION SYSTEM**

AKITA WEBSITE

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AKITA Description

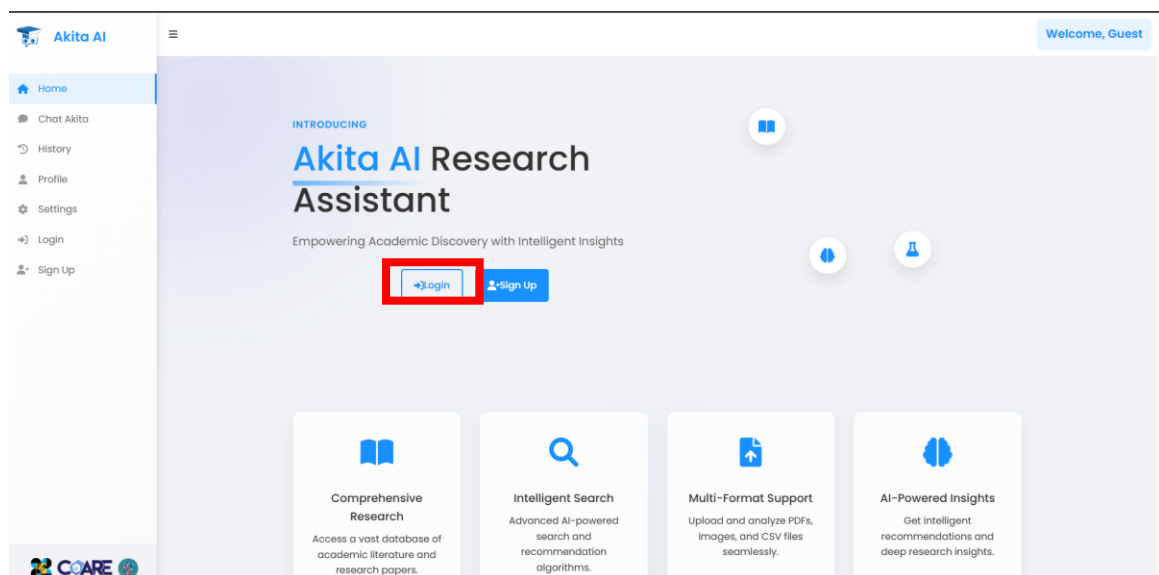
AKITA, an AI-powered literature recommender system at TUP Manila. Utilizing a secure High-Performance Computing (HPC) cluster, the system functions as a research assistant, linking datasets to past studies, with a focus on ECE theses from 2014 to 2024. In addition, it includes IEEE Xplore articles, categorized for relevance to support comprehensive literature reviews.

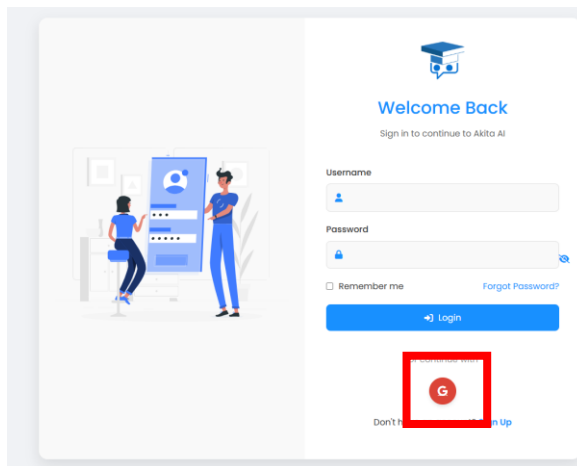
AKITA Purpose

The AKITA website aims **to support TUP ECE researchers in conducting their Review of Related Literature (RRL)**, with a particular focus **on locally-produced research**. It also helps users identify whether their research ideas have already been explored by previous ECE students, promoting originality and informed topic selection.

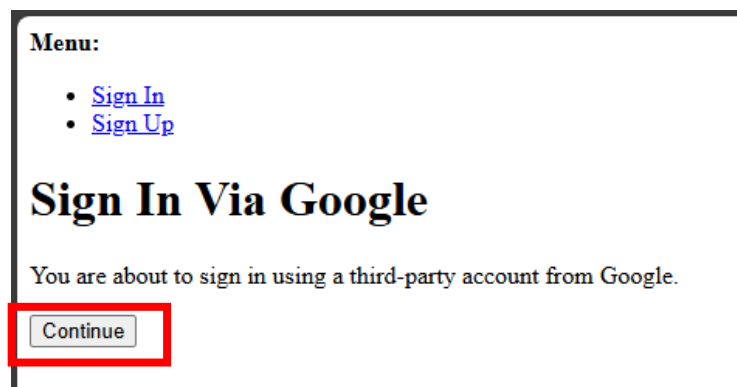
Website Registration and Access

1. Open the AKITA Website via your preferred browser. Please take note that you can access AKITA in IRTC Building.
2. Press **Login** > then press the **Google Icon**.



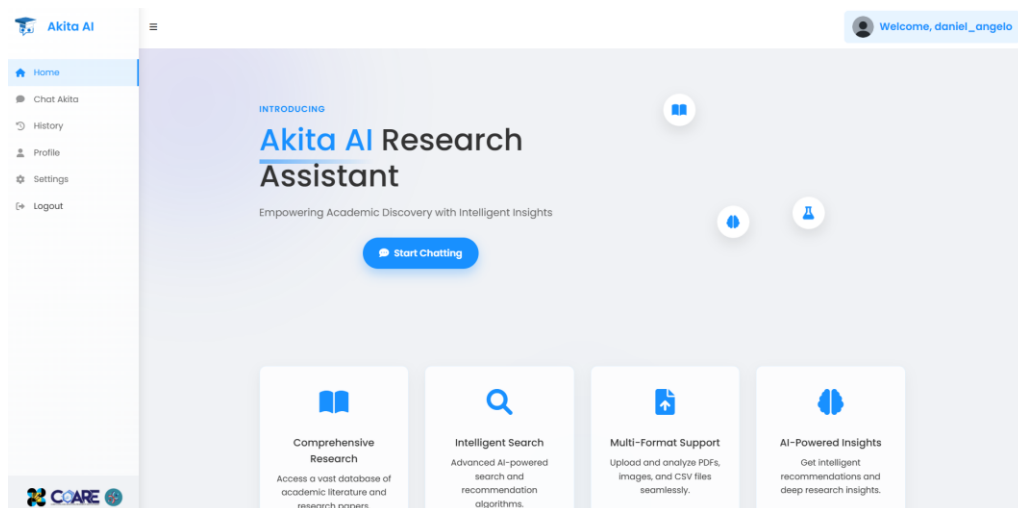


3. Then press the “**continue**” button.



4. Then sign in your personal / TUP account.

5. After that, **you are now registered!** You will see in the website if you are totally registered by having your username.

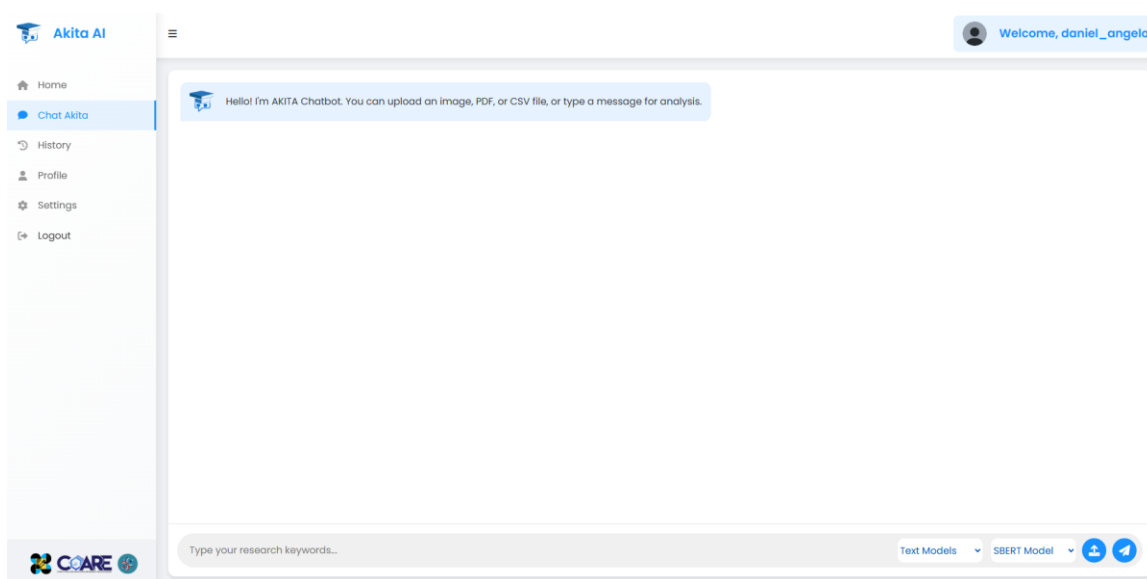


6. If you want to login again, use your username and password.

AKITA Features

1. Chat Feature

The AKITA Chat Feature enables researchers to search for relevant topics using either **text** or **image** inputs. Each input type is powered by two distinct AI models, giving users the flexibility to choose the model that best suits their needs.

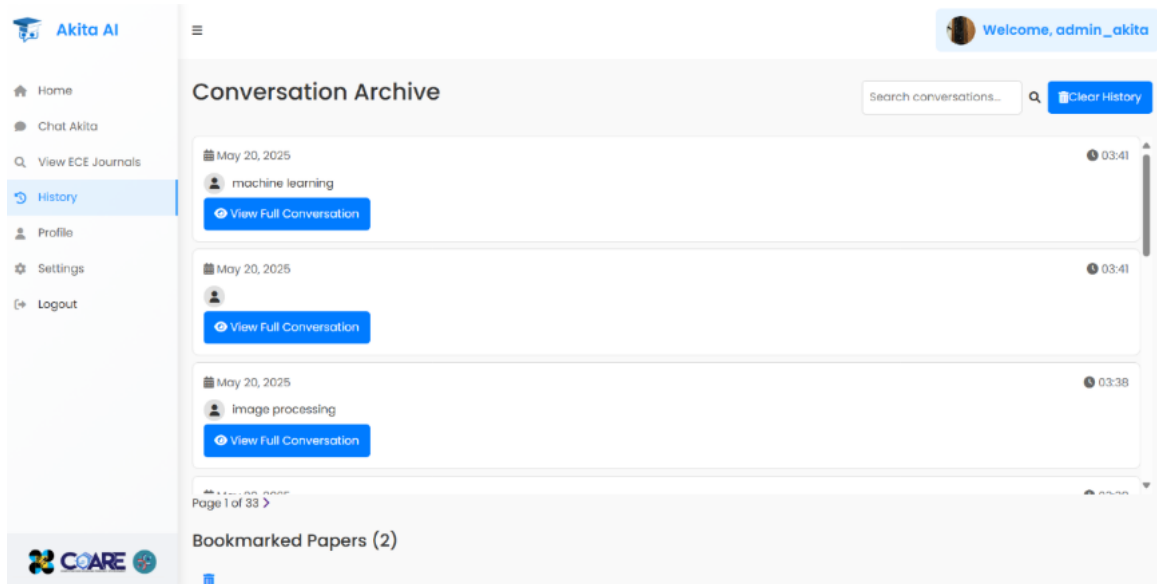


2. View Journal Feature

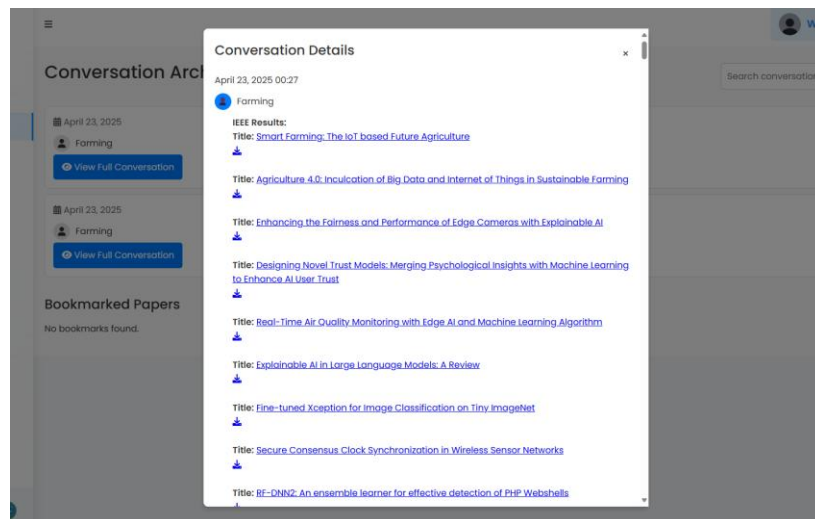
In this page, you can search specific paper based on our database. You can also filter them based on title and categories.

3. History Feature

You can revisit your previous queries using the History Feature.

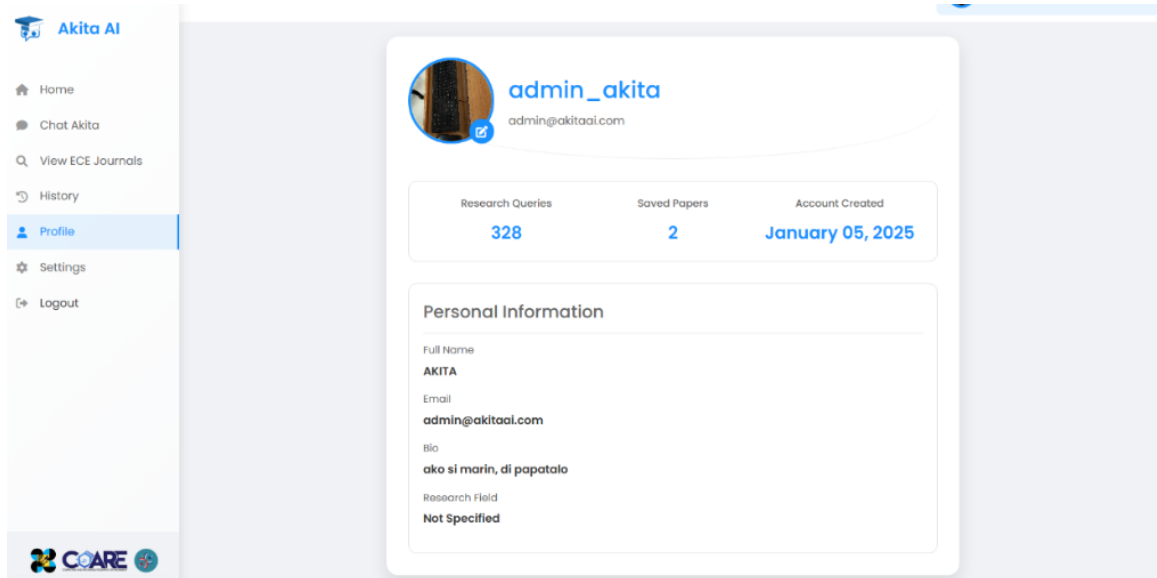


By pressing the “View Full Conversation” button, a window will appear where you can see the recommended papers for your query.



4. Profile Feature

The **Profile Page** allows you to see relevant stats such as shown.



5. Settings Feature

The **Settings Page** allows you to update your profile information. You can change your password and update your profile picture.

Profile Settings
Manage your account and personalize your experience

admin_akita
admin@akitaai.com

Username: admin_akita

Email Address: admin@akitaai.com

Full Name: AKITA

Profile Image: Choose file

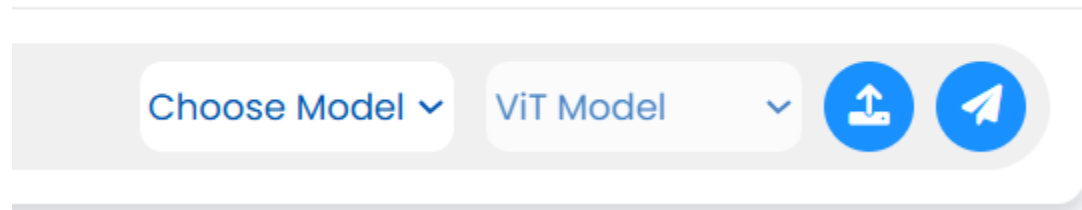
Bio: ako si marin, di papatalo

Update Profile

How to Prompt

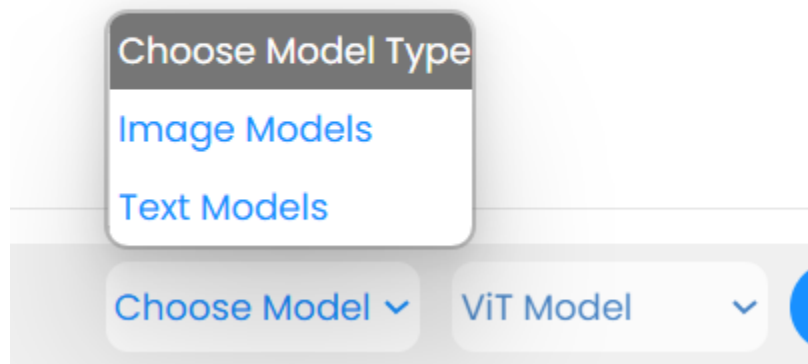
1. Choose a model

Before sending or uploading a prompt, select the model you want to use.



2. Choose the model type

Decide whether you want to use a **text** model or an **image** model.



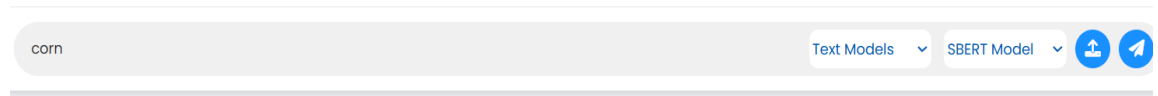
3. Select a specific model

After choosing the model type, select the specific model you want to use.

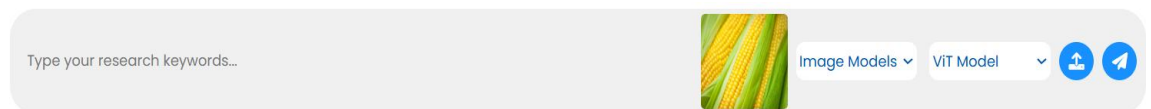


4. Prepare your prompt

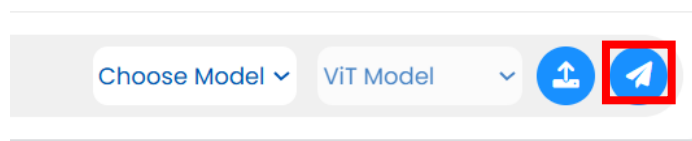
- If you're using a **text model**, type your prompt. Take note that you can type keywords, phrases or sentences that is related to your topic.



- If you're using an **image model**, upload an image as your prompt.



5. **Click Send Button.**



Text Models

- **BERT (Bidirectional Encoder Representations from Transformers):** A powerful transformer-based model designed for understanding the context of words in a sentence. BERT performs well in general language understanding tasks.
- **SBERT (Sentence-BERT):** An extension of BERT fine-tuned specifically for producing sentence embeddings. SBERT is optimized for semantic similarity tasks, making it ideal for identifying closely related research topics based on full-sentence queries.

Image Models

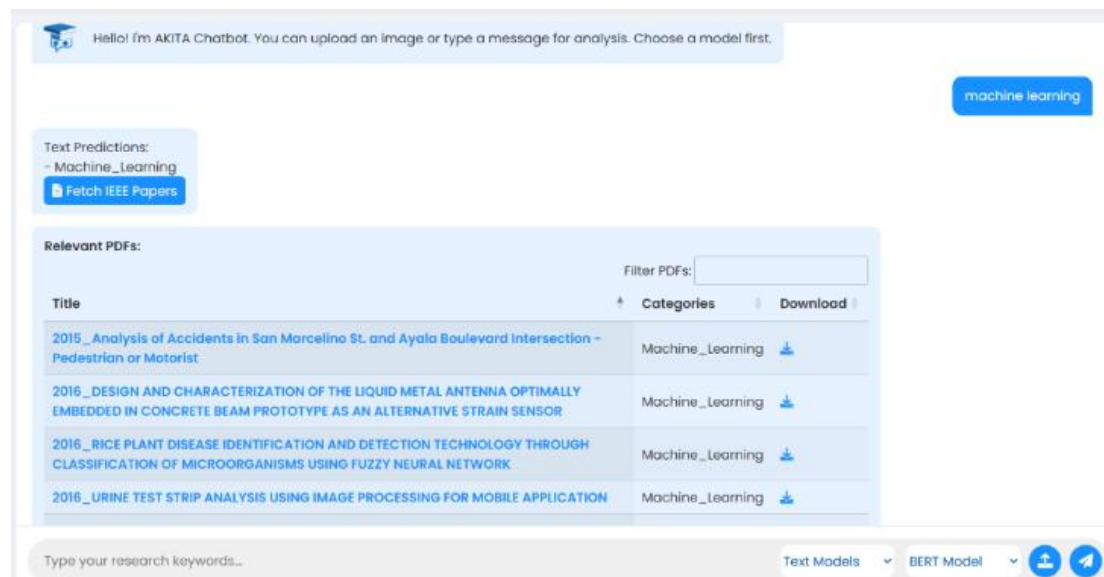
- **ViT (Vision Transformer):** A model that treats an image as a sequence of patches and applies transformer-based attention mechanisms to

understand visual features. It excels at capturing global patterns and structures within an image.

- **ResNet-50 (Residual Network):** A deep convolutional neural network that focuses on learning low- to mid-level image features. It is effective for standard image classification tasks and has proven robust in many computer vision benchmarks.

Query Results

- You can also fetch relevant papers from the TUP papers or IEEE Xplore.
- On every query you gave, it will return downloadable relevant journals/manuscripts that are in PDF format.



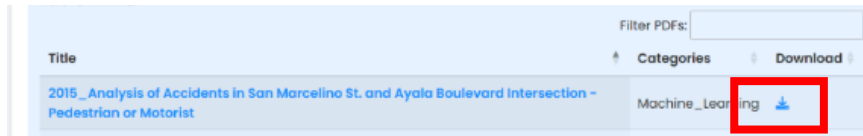
Disclaimer

All AI models used in AKITA are trained on **open-source datasets** and are tailored to classify inputs based on predefined **research categories**. Each search query—whether text or image—is **categorized** by the selected model, and predictions are returned based on the **most relevant category**.

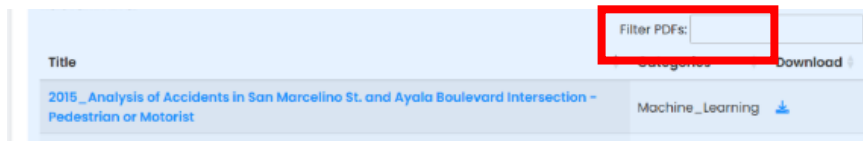
As a result, the output may not always directly match the specific keyword used in the search.

For example, a query like “fish” may be categorized under **Agriculture**, and the system will return research journals within that domain, even if the exact term "fish" does not appear prominently in the results.

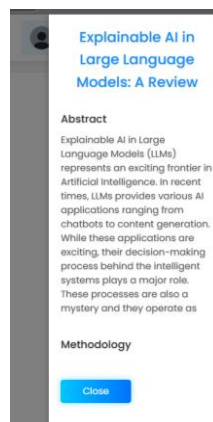
- You can download the recommended journal to view its contents.



- You can filter the results based on categories or titles.



- Pressing the **Title** will prompt a pop-up slide where you can see the Abstract and Title of the paper.



Contact Support

- Only authorized personnel or the proponents themselves should perform software repairs or modifications on the website. Unauthorized actions may void the service agreement or cause further damage.
- If you encounter any issues, bugs, or difficulties with the website, please email the team at akita.tupm@gmail.com.